VECTOR7: Zero-Trust Claim-Level Credibility Assessment for Conversational AI

Anonymous Supplementary Material for Review

Appendices A–C and detailed claim-level audit outputs

Scope. This anonymous supplement contains supporting material referenced by the main manuscript: the claim-type/probe-applicability matrix (Appendix A), individual probe vectors before and after consistency conditioning (Appendix B), and detailed claim-level interpretation tables (Appendix C). The supplement is intended to support transparency and reproducibility; the main manuscript contains the complete framework definition, algorithms, aggregate results, and conclusions required to understand the paper.

Appendix A: Claim Types and Probe Applicability

VECTOR7 probes are diagnostic dimensions whose applicability and normative significance depend on the type of claim being evaluated. Table A.1 specifies representative claim types and which probes are required, optional, or typically not applicable. Indiscriminate probe application can introduce unnecessary non-applicable outcomes and dilute diagnostic power. Mapping claim types to relevant probes reduces analytic overwork, supports consistent handling of skipped probes, and enables fair comparison across evaluations.

Table A.1: Applicability of VECTOR7 Probes to Representative Claim Types

Claim Type	V1 Procedural Integrity	V2 Contextual Credibility	V3 Operational Validity	V4 Semantic Stability	V5 Contextual Robustness	V6 Evidentiary Grounding	V7 Self-Audit
Factual Claims	O	O	O	R	O	R	O
Procedural Claims	R	O	R	O	O	O	R
Technical/System Claims	R	R	R	O	O	R	R
Reasoning Claims	O	O	R	R	R	O	O
Safety/Policy Claims	O	O	O	R	R	O	R
Evidentiary Claims	O	O	O	O	O	R	R
Hypothetical/Counterfactual Claims	R	O	R	R	R	—	O

Legend: R = required/strongly recommended; O = optional/situational; — = typically not applicable.

Probe applicability reflects the epistemic obligations inherent in different claim types. V6 (External Evidence) is normatively central for factual or evidentiary claims but is often intentionally inapplicable to hypothetical or counterfactual claims. Reasoning claims primarily rely on V3–V5, which assess task execution, semantic invariance, and contextual transfer, while technical/system claims emphasize V1–V3 and V6 to evaluate mechanisms, reproducibility, and external grounding. In safety/policy claims, V7 (Self-Audit) and V5 (Context/Transfer) are particularly important because they probe uncertainty recognition and consistent behavior across contexts.

Weighting clarification. A probe that is structurally inapplicable to a claim type receives zero normative weight, and the weights of applicable probes are renormalized to sum to one. An applicable probe that is intentionally omitted retains its normative weight and therefore reduces coverage. This distinguishes genuine non-applicability from under-testing.

Appendix B: Result of Individual Probes

Table B.1 reports the Independent Credibility Vector (ICV) for each claim before cross-probe consistency conditioning. Table B.2 reports the corresponding Consistency-Conditioned Credibility Vector (CCV);

highlighted IN cells indicate PASS outcomes conservatively downgraded to INCONCLUSIVE after a cross-probe contradiction or commitment drift was detected.

Tables B.1–B.2: Independent and Consistency-Conditioned Credibility Vectors

	Question	Ans/Original Claim	V1- Procedural Integrity	V2- Contextual Credibility	V3- Operational Validity	V4- Semantic Stability	V5- Contextual Robustness	V6- Evidentiary Grounding	V7- Self-Critique and Confidence
Gemini	1	Yes	1	1	1	0	1	0	1
	2	Yes	1	1	1	0	0	1	1
	3	No	1	1	1	1	1	1	1
	4	No	1	1	1	1	0	1	1
	5	No	1	1	1	1	0	1	1
	6	No	1	1	1	1	1	1	1
	7	Yes	1	1	1	1	1	1	1
	8	No	1	1	1	1	1	1	1
	9	No	1	1	1	1	1	1	1
DeepSeek	1	Yes	0	0	1	0	1	1	1
	2	Yes	0	0	0	0	1	1	1
	3	Yes	0	0	1	0	1	1	1
	4	No	1	1	0	1	0	1	1
	5	Yes	0	0	1	1	1	1	1
	6	No	1	1	1	1	1	1	1
	7	No	1	1	0	1	1	1	1
	8	No	1	1	0	1	1	1	1
	9	Yes	1	1	0	0	1	1	1

	Claim	V1.Procedural Integrity	V2- Contextual Credibility	V3- Operational Validity	V4- Semantic Stability	V5- Contextual Robustness	V6- Evidentiary Grounding	V7- Self-Critique and Confidence
Gemini	T1 – Yes	IN	IN	1	0	1	0	IN
	T2 – Yes	IN	IN	1	0	0	IN	IN
	T3 – No	1	1	1	1	1	1	1
	T4 – No	IN	1	1	IN	0	1	1
	T5 – No	1	1	1	1	0	1	1
	T6 – No	1	1	1	1	1	1	1
	T7 – Yes	1	1	1	1	1	1	1
	T8 – No	1	1	1	1	1	1	1
	T9 – No	1	1	1	1	1	1	1
DeepSeek	T1 – Yes	0	0	1	0	1	IN	1
	T2 – Yes	0	0	0	0	1	IN	1
	T3 – Yes	0	0	1	0	1	IN	1
	T4 – No	1	1	0	1	0	1	1
	T5 – Yes	0	0	1	1	1	IN	1
	T6 – No	1	1	1	1	1	1	1
	T7 – No	IN	IN	0	1	1	IN	1
	T8 – No	1	1	0	1	1	1	1
	T9 – Yes	1	1	0	0	1	1	1

Appendix C: Credibility Decision Interpretation Layer

Tables C.1 and C.2 present the claim-level VECTOR7 outputs for Gemini and DeepSeek, including the quantitative metrics (CES, CCM, CEE, CSR), the resulting credibility decision, a concise explanation of the decision process, and the specific epistemic strengths or weaknesses identified from the conditioned probe outcomes. The interpretation layer traces the V7-EDA and V7-CPCC decision flow and converts raw scores into an auditable explanation of why a claim is classified as Credible or Inconclusive.

For fully decisive probe outcomes, the seven PASS/FAIL dimensions yield $2^7 = 128$ binary epistemic patterns. VECTOR7 additionally preserves INCONCLUSIVE and SKIPPED outcomes explicitly in the ICV and CCV rather than collapsing them into the binary state taxonomy. The detailed explanations below therefore complement, rather than replace, the underlying probe vectors.

Interpretation is human-facing and operational: a scalar label alone is insufficient in high-stakes settings. The decision-process explanation states how coverage, epistemic strength, instability, and consistency produce the final outcome, while the diagnostic column identifies the probe-level strengths and weaknesses that an analyst can inspect or remediate.

Table C.1: Credibility Analysis and Results for Gemini Claims

Claim	CES	CCM	CEE	CSR	Decision	Explanation of Decision Process	Analysis of Specific Epistemic Strength(s)/Weakness(es)
T1 – Yes	0.50	0.57	High	0.40	Inconclusive	Coverage below threshold and high instability after conditioning	Operational validity and limited contextual robustness are present, but semantic drift and lack of evidentiary grounding dominate; procedural integrity, contextual credibility, and epistemic awareness are under-established.
T2 – Yes	0.33	0.43	High	0.20	Inconclusive	Insufficient coverage and weak epistemic strength	Operational validity is present, but semantic stability and contextual robustness fail; procedural integrity, contextual credibility, evidentiary grounding, and epistemic awareness remain under-tested.
T3 – No	1.00	1.00	0	1.00	Credible	Full coverage, unanimous support, no instability	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all satisfied.
T4 – No	0.80	0.71	Moderate	0.67	Credible	Meets coverage and strength thresholds; instability below override	Contextual credibility, operational validity, evidentiary grounding, and epistemic awareness are strong; contextual robustness fails; procedural integrity and semantic stability are not decisive.
T5 – No	0.86	1.00	Low	1.00	Credible	Full coverage with strong pass dominance	Procedural integrity, contextual credibility, operational validity, semantic stability, evidentiary grounding, and epistemic awareness hold; contextual robustness is limited.
T6 – No	1.00	1.00	0	1.00	Credible	Full coverage, full strength, no instability	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all satisfied.
T7 – Yes	1.00	1.00	0	1.00	Credible	Full coverage and consistent support	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are consistently satisfied.
T8 – No	1.00	1.00	0	1.00	Credible	Full coverage and no detected instability	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all present.
T9 – No	1.00	1.00	0	1.00	Credible	Full coverage and coherent probe outcomes	Procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness are all satisfied.

Table C.2: Credibility Analysis and Results for DeepSeek Claims

Claim	CES	CCM	CEE	CSR	Decision	Explanation of Decision Process	Analysis of Specific Epistemic Strength(s)/Weakness(es)
T1 – Yes	0.50	0.86	High	1.00	Inconclusive	Coverage is sufficient, but weak strength and high instability trigger conservative downgrade	The claim exhibits operational validity and partial contextual robustness, but suffers from procedural void, contextual incoherence, and semantic drift. Evidentiary grounding is under-determined, while epistemic awareness is present, yielding unstable support.
T2 – Yes	0.33	0.86	High	1.00	Inconclusive	Adequate coverage but insufficient epistemic strength	The claim shows contextual robustness and epistemic awareness, but fails with procedural void, contextual incoherence, operational failure, and semantic drift, overwhelming its limited strengths.
T3 – Yes	0.50	0.86	High	1.00	Inconclusive	Mixed probe outcomes produce instability despite sufficient coverage	The claim maintains operational validity and contextual robustness, but is undermined by procedural void, contextual incoherence, and semantic drift. Evidentiary grounding remains under-determined.
T4 – No	0.71	1.00	Moderate	1.00	Inconclusive	Full coverage achieved, but pass dominance falls below threshold	The claim demonstrates procedural integrity, contextual credibility, semantic stability, evidentiary grounding, and epistemic awareness, but fails due to operational failure and contextual collapse, limiting executability and transfer.
T5 – Yes	0.67	0.86	Moderate	1.00	Inconclusive	Coverage sufficient, but strength below credibility threshold	The claim shows operational validity, semantic stability, contextual robustness, and epistemic awareness, but exhibits procedural void and contextual incoherence, weakening foundational support.
T6 – No	1.00	1.00	0	1.00	Credible	Full coverage, full strength, no instability	The claim satisfies procedural integrity, contextual credibility, operational validity, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness with no detected failure mode.
T7 – No	0.75	0.57	Low	0.67	Inconclusive	Coverage below threshold despite reasonable strength	The claim demonstrates semantic stability, contextual robustness, and epistemic awareness, but fails with operational failure. Procedural integrity, contextual credibility, and evidentiary grounding remain under-determined.
T8 – No	0.86	1.00	Low	1.00	Credible	Full coverage and strong pass dominance	The claim exhibits procedural integrity, contextual credibility, semantic stability, contextual robustness, evidentiary grounding, and epistemic awareness, with a localized operational failure.
T9 – Yes	0.71	1.00	Moderate	1.00	Inconclusive	Full coverage but insufficient epistemic strength	The claim shows procedural integrity, contextual credibility, contextual robustness, evidentiary grounding, and epistemic awareness, but is undermined by operational failure and semantic drift, limiting credibility.